

Aditya Rao

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EDUCATION

McMaster University

Hamilton, ON

B.Eng in Software Engineering (CO-OP)

Sept 2023 – Present

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Digital Systems, OOP, Discrete Math, Software Design, DSA, Computer Architecture

EXPERIENCE

Software Developer Intern

May 2025 – Aug 2026

IBM

Markham, ON

- Built an AI diagnostic platform using IBM Bob to triage Liberty build breaks with 99% accuracy across 100 test failures, reducing incident investigation time by 2 days per break.
- Led modernization of legacy engineering support tools (EJB/Servlets, Ruby on Rails) to React, Spring Boot, Quarkus, and OpenLiberty microservices, completing a multi-year migration of mainframe based ticketing system to Salesforce.
- Architected a work-distribution analytics app for 400+ developers to track engineering time across CVEs, features, and QA, replacing shared spreadsheets to save 1 day a week of overhead per manager.
- Accelerated WebSphere CI/CD reliability by refactoring 20+ legacy FAT test suites into modular JUnit frameworks.
- Diagnosed 50+ multi-platform build failures across Linux, macOS, Windows, and z/OS while maintaining core WebSphere Application Server, WebSphere Liberty, and contributing to OpenLiberty as an open source developer.

Full Stack Developer Intern

May 2024 – Aug 2024

Medium AI

Hamilton, ON

- Engineered real-time doctor-patient voice-based transcription with gRPC and Firebase, achieving sub-500ms streaming latency by load balancing OpenAI Whisper across self-hosted and managed instances.
- Integrated AI diagnostic hooks to analyze transcripts, highlighting clinical insights for attending physicians.
- Developed and maintained React and TypeScript website frontend for Medium AI, contributing to 2 first-place and 1 second-place finishes in the McMaster Startup Survivor competition, and helping secure \$10,000 in funding.

PROJECTS

BYoAI Deployment Platform | 🌐 | *Go, TypeScript, Docker, MCP, OpenTelemetry*

May 2026 – Present

- Architected a multi-tenant AI agent orchestration platform in Go and TypeScript with MCP tool-calling, subagent hierarchies, and OpenTelemetry.
- Built a containerized agent execution engine with Docker for dynamic worker allocation, security isolation, and pluggable storage systems.
- Developed a multi-provider LLM abstraction supporting OpenAI, Claude, Gemini, and self-hosted models across REST and event-driven pipelines.

WebCraft | 🌐 | *C++, CMake, Vcpkg, Asynchronous Programming, Github Actions*

Apr 2025 – Sept 2025

- Engineered an asynchronous programming library in C++ 23, featuring coroutine-based task management, concurrent streams, and high-performance networking/file I/O modules to simplify cross-platform async development for C++ developers.
- Automated testing and release workflows using gtest (130+ tests), GitHub Actions CI/CD, and published a custom vcpkg port for standardized cross-platform distribution.

Resume Builder | 🌐 | *Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker*

Aug 2024 – Feb 2025

- Built an AI-powered resume builder with Next.js, Spring Boot, and FastAPI for real-time editing and version control.
- Integrated LangChain to automate ATS keyword optimization, intelligent scoring, and template tailoring.
- Accelerated resume tailoring workflow by ~90%, cutting manual editing time from ~2 hours to <10 minutes per application.

FashioNova | 🌐 🌐 | *Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask*

Oct 2024 – Oct 2024

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

Smart Study Buddy | 🌐 🌐 | *Python, Flask, Wolfram Alpha API, OpenAI API, Next.js, Typescript*

Oct 2024 – Oct 2024

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

QuakeGuard | 🌐 🌐 | *Next.js, Flask, Typescript, Python, TensorFlow, Keras, SciKit Learn, Pandas*

Jan 2024 – Jan 2024

- Developed an earthquake prediction tool using ML models showcased at DeltaHacks X and GDSC McMaster
- Visualized global earthquake data and high-risk areas with Next.js and Python, Flask, SciKit Learn, and TensorFlow

LEADERSHIP

Sys-Admin and Fullstack Developer Executive

Nov 2024 – Present

IEEE McMaster Student Branch

Hamilton, ON

- Set up an RPi 3B+ Cluster to handle up to 100 concurrent clients with <50ms response time
- Managed and provisioned a CI/CD pipeline which generates Docker images for our Kubernetes cluster
- Developed IEEE McMaster student branch website using Next.js, leveraging GitHub Projects and Issues as the admin dashboard for managing events and member contributions

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C#, C++, Golang, SQL, C, Verilog
Frameworks: Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask, UWP
Developer Tools: AWS, GCP, Git, Docker, Kafka, OpenShift, Kubernetes, Redis, Infinispan
Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, JavaFX, Redux, JQuery, React, Mongoose
Hobbies: Sketching, Painting, Violin (Western and Carnatic), Badminton, Swimming, Chess

CERTIFICATIONS

Clean Code – UdemylSeptember 2025

IBM Agile Explorer – IBMSeptember 2025

The DevOps Essentials - The Handbook – UdemylSeptember 2025

Introduction to Containers, Kubernetes, and OpenShift – IBMAugust 2025

Practical OpenShift for Developers - OpenShift 4 – UdemylAugust 2025

[NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2025 – UdemylFebruary 2025

Docker & Kubernetes: The Practical Guide [2025 Edition] – UdemylJanuary 2025